

Awareness of the real-world environment when using augmented reality head-mounted display

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ABSTRACT

Augmented reality (AR) systems are becoming common tools in industrial workplaces. However, factory workers are still concerned about whether head-mounted display (HMD)-based AR systems distract their awareness of the environment and therefore pose safety risks. The purpose of this study was to assess users' experience of real-world awareness when using an AR system. 19 study participants played a wooden block logic game in a laboratory with three different setups: real, AR and virtual reality (VR). Based on this study, it can be concluded that HMD-based AR systems do not decrease users' awareness of their surroundings if the virtual content is minimal and the task is done while seated. However, it was seen that more research in this area with more interactive virtual content is required. This study is an important step in understanding how AR may affect future work in industrial and safety-critical environments.

1. Introduction

Augmented reality (AR) technologies are becoming mature enough to be adopted in everyday use. Pierdicca et al. (2017) see AR as one pillar of new industrial environments. However, there are still challenges to overcome when implementing AR systems in industry (Masood and Egger, 2019; Palmarini et al., 2018). For example, employees are concerned about the use of head-mounted display (HMD)-based AR systems because they may distract them (Aromaa and Kaasinen, 2018; Sunwook et al., 2016). They are afraid that the AR systems could decrease workers' awareness of the surrounding environment. The concern is valid in the sense that, for example, the use of the Pokémon Go game has led to accidents. Barbieri et al. (2017) point out that these mobile games could pose a significant risk to users if they play while walking, cycling, or driving, because the users become distracted and may ignore other traffic and surrounding hazards. In addition, Kim and Gabbard (2019) found out that AR head-up displays (HUDs) can direct the vehicle driver's attention away from critical road elements (e.g., other vehicles or road signs). However, they conclude that the AR content can be informative or distractive based on the selected perceptual forms of graphical elements.

By definition, AR allows the user to see the real world, with virtual objects superimposed upon or composited with the real world (Azuma

et al., 2001). The main application areas of AR technologies are in design, maintenance and assembly (Nee et al., 2012). Nee et al. (2012) categorize AR-assisted systems into three groups based on the information presentation method: presenting information on HMD (see examples Helin et al., 2018; Henderson and Feiner, 2009; Makris et al., 2016), presenting information with a tablet or smartphone (see examples Aaltonen et al., 2016; Aromaa and Väänänen, 2016; Polvi et al., 2018; Sánchez et al., 2016), and projecting information at the workplace (see examples Doshi et al., 2017; Funk et al., 2017; Korn et al., 2013). In this introduction, also some studies related to smart glasses and HUDs are addressed even though they are not always actual AR because the information is merely "glued" on the display to the view. In actual AR systems, the digital information is presented as if it exists in reality.

Attention, awareness and cognitive load are issues that are important to consider when using AR systems on a factory floor because they can have an effect on safety. When people are disrupted by divided attention, it is possible that they experience inattention blindness. Inattention blindness results from a lack of attention that is not associated with vision defects or deficits, as an individual fails to perceive an unexpected stimulus in plain sight (e.g. the famous gorilla study by Simons and Chabris, 1999). Situation awareness (SA) can be defined as a perception of elements in the environment within a volume of time and space, the comprehension of their meaning and the projection of their

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future status (Endsley, 1988). In other words, SA means that people are aware of the specific information that is needed to perform a task. Context awareness is similar to SA, but in context awareness users are aware of all relevant information a specific environment can provide to them.

Some AR studies have proven that AR systems, smart glasses and HUDs can have a positive effect on users' context awareness, safety and/or decrease cognitive load. For example, when increasing context awareness users can be provided with information otherwise not seen in their environment: Schwarz and Fastenmeier (2018) studied AR use when providing warnings about obstructed hazards and Grubert et al. (2017) discussed sensing a user's current context. Park and Park (2019) agree in their review that automotive HUDs can often improve driving safety and enhance the driver's SA. Borgmann et al. (2017) studied the feasibility, safety and usefulness of Google Glass for urological surgery. It was seen that smart glass-assisted urological surgery is both feasible and safe and additionally provides several useful functions for urological surgeons. Kim and Dey (2009) compared two car navigation systems and one of them was an AR system. They found that the AR system was useful, especially for older drivers. In addition, the drivers using the AR display system had significantly fewer navigation errors and divided attention-related issues. According to Kim et al. (2019) having graphic-based information constantly visible on the HMDs led to increased job performance and reduced perceived workload as compared to the use of text-based information, or the use of a traditional paper pick list. Schwarz and Fastenmeier (2018) conclude that for warning situations while driving a car well designed AR provide the opportunity to increase the amount of information conveyance without distracting the driver. Rowen et al. (2019) studied Google Glasses when navigating marine vessels. They agreed that the use of wearable AR displays could improve operator performance and situation awareness.

There are also studies that indicate that AR systems may increase the cognitive load. According to Gross et al. (2018), the cognitive workload on forklift drivers increases significantly when using display information devices such as HMDs. Wang et al. (2019) study shows that users experience more workload when using Google Glass-based AR instructions in disassembling a smartphone than printed manual. Bau-meister et al. (2017) evaluated cognitive load of three AR display technologies: projector-based display (spatial AR), Microsoft HoloLens and Samsung Gear VR with monitor instructions. They discovered that a limited field of view could introduce increased cognitive load requirements. The findings suggest that some of the inherent restrictions of HMDs (e.g., field of view, resolution, vergence, accommodation) materialize as increased user cognitive load. Kerr et al. (2012) found that divided attention between the AR display and the environment brought up some safety concerns. At least four (out of nine) participants failed to notice they were approaching a road junction, and had to be warned of the approaching traffic. Kim et al. (2019) agrees that walking while processing visual information presented via HMDs may have important workplace safety implications. Kruijff et al. (2010) indicated that perceptual problems that may occur in AR can derive from issues in the environment (e.g. visibility), capturing (e.g. colour contrast), augmentation (e.g. occlusion), display device (e.g. field of view) and/or the user (e.g. individual differences). According to Aaltonen and Laarni (2017), smart glasses can be problematic from the aspects of comfort and distraction. For example, participants said that the smart glasses force the users to divide their attention among multiple things.

How much AR systems distract users and make them unaware of what is happening in their environment seems not to have been studied widely. Therefore, there is a need to study especially the user experience of HMD-based AR systems more thoroughly. According to Kim et al. (2019) limited information is available regarding the effective use of head-worn displays in workplaces. In addition, the studies are dispersed in the field of science, and are related to several different kinds of tasks (e.g. driving, surgery, assembly). There is considerable variation in available technologies: many studies consider head-worn displays

(monocular, binocular) and not the actual AR HMDs where virtual objects are presented as a part of a real environment.

Our study addresses the gap in the research concerning how much HMD-based AR systems distract users. The goal of the study was to investigate how users experience their real-world awareness while using an AR system. The study was conducted in a laboratory and the participants performed the same tasks with three different systems (real, AR and virtual reality (VR)) on the reality-virtuality (RV) continuum (Fig. 1). The RV continuum is a continuous scale categorising displays from the completely real to the completely virtual (Milgram et al., 1995). By studying three different reality-virtuality level systems it is possible to understand how users experience the AR system compared to real and virtual systems. Kalawsky (1993) defines VR as a technology which provides human beings with the means of manipulation and sensory modalities in a virtual environment. Practically, it means that humans are able to navigate in the virtual environment (e.g. move from one place to another), manipulate objects (e.g. pick up a virtual object), and get sensory feedback (e.g. see and hear things).

This paper raises the important research topic of how distracting AR systems can be. The results of this study can be used when studying the suitability of HMD-based AR systems for industry use. In addition, the study highlights the need for improved evaluation methods for studying users with AR systems. The structure of the paper is as follows: Section 2 presents material and methods, Section 3 describes results of the study, Section 4 discusses the findings and proposes directions for future research, and Section 5 summarizes research conclusions.

2. Material and methods

This section describes the study's design, participants, three test setup systems, data collection methods, the test procedure and data analysis methods.

2.1. Study design

The main goal was to study users' real-world awareness when using HMD-based AR system. An experimental approach (Sanders and McCormick, 1993) was adopted for this study. A within-subject design was used. The independent variable was the virtuality level. The dependent variable was the experience of the system. There were three setups for the virtuality level: real, VR and AR. Related research and our discussions with industry stakeholders have shown that there is a concern that the HMD-based AR systems could distract the worker. Therefore, we assume:

H1. The HMD-based AR system decreases the users' real-world awareness.

2.2. Participants

Two participant groups were used in this study: users and experts. Nineteen participants (13 male and 6 female) took part in the user study. The participants were recruited via an announcement posted at the University and the Research Institute. Participants got two movie tickets as compensation for taking part in the study. The average age was 38 and the age range was 21–57. There were 10 research scientists, 7 university students, 1 quality specialist and 1 research technician. Two of the participants were active users of AR technologies, 9 had some experience, 6 were familiar with the term and 2 did not have knowledge or earlier experience. 12 participants had some experience of VR technologies and 7 knew the term. 10 participants had eyeglasses. 2 kept their eyeglasses on while using the VR system and 4 kept their eyeglasses on while using the AR system. It was ensured that all users were able to see the displayed information.

11 experts (10 Male and 1 female) answered the AR/VR expert questionnaire. The web-based questionnaire was distributed to Forums

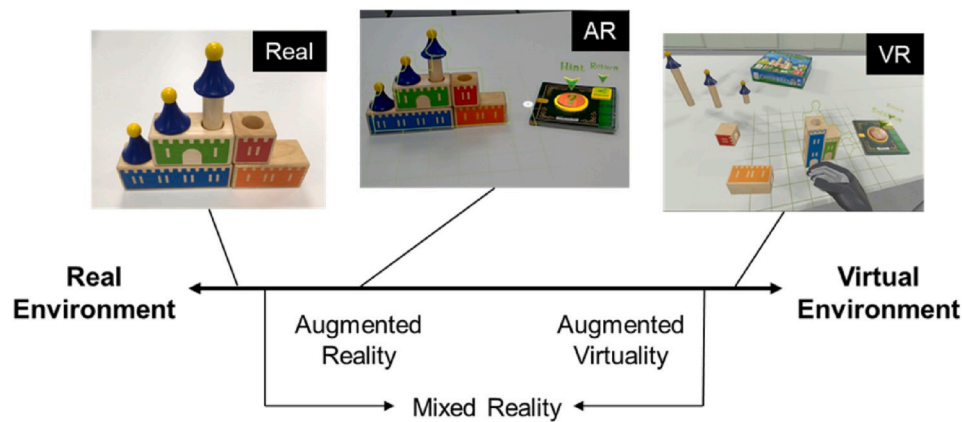


Fig. 1. Three different study setups (real, augmented reality and virtual reality) on the (Milgram et al., 1995)'s reality-virtuality continuum.

(e.g. EuroVR mailing list, EuroVR LinkedIn and EuroVR Facebook) and via email to known AR/VR experts by the researchers. The average age was 45 and the age range was 28–61. There were 8 research scientists, 2 engineers and 1 PhD student. Five had used AR technologies actively 0–5 years, one 11–15 years and five over 16 years. Seven participants had used VR technologies actively 0–5 years and four over 16 years.

2.3. Systems descriptions

This section describes the used systems in detail. The systems are introduced in the same order in which they were used during the user test (real, VR, AR). As a basis for all three setups, a game called Castle Logix was used. The game was selected for the task because playing the game requires concentration and it was easy to replicate for all setups. In addition, building a castle has similar features as an assembly task in industry. The view of the user in real, VR and AR setups can be seen in Fig. 2.

2.3.1. Real system

The real system was a game called Castle Logix (SmartGames). The purpose of the game is to build castles based on the instructions in the book using four wooden blocks and three wooden towers. During the game, only master level instructions were used. For the master level, only the castle outlines are shown, but not the colours of the blocks (Fig. 2). The challenge of the game is to work out the correct order and orientation of the blocks to produce the depicted silhouette and stable structure. The participants were sitting in a meeting room chair and played the game on the top of the table (Fig. 3).

2.3.2. Virtual reality system

When using the VR system, the participants were seated in the same physical room as in the real setup (Fig. 3), but all activities such as building the castle were done with virtual objects in the virtual version of the room (Fig. 2, top right). The participants wore the HTC Vive Pro HMD (Fig. 4), which shows the virtual surroundings and the virtual

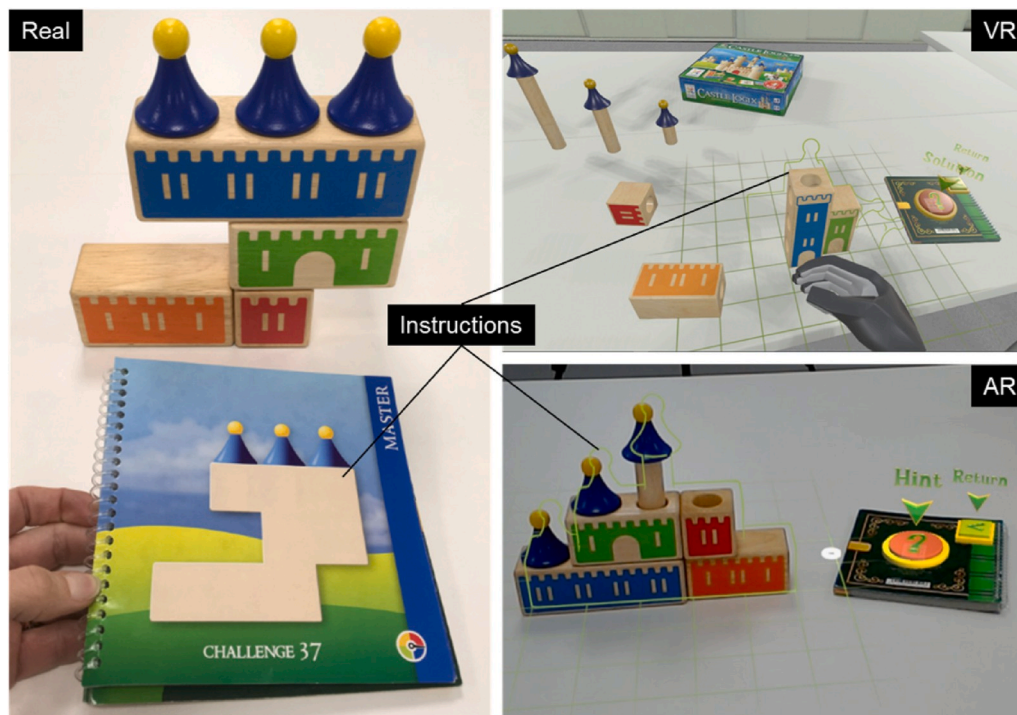


Fig. 2. Participants' perspectives of the different study setups: real system on the left, virtual reality (VR) system on the top right and augmented reality (AR) system on the bottom right. In the VR and AR setups, the instructions are shown as the faint green castle outlines (the grid is used for showing the gaming surface). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)



Fig. 3. Three test setups for the virtuality level: (1) real, (2) virtual reality and (3) augmented reality. In the virtual reality setup (middle), the participant is manipulating virtual objects shown on the HMD, and the surrounding real world is not visible to the participant.

blocks (Fig. 2), the participant’s hands were tracked with a Leap Motion device attached on the HMD and they were able to see their own virtual hands. The participant’s surroundings in the virtual environment simulated the real room, including the participant’s chair and table.

For describing the VR system in detail, a systematic approach (Aromaa, 2017) was adopted (Table 1). The VR system includes 3D models (static model characteristics) of castle blocks and towers, book, grid, buttons on the book, instruction texts, solutions for four castles, hands, Castle Logix game box, and room with tables, chairs, walls, windows and cabinets. The VR system includes dynamic features (dynamic model characteristics): participants are able to use their virtual hands in the virtual environment and manipulate virtual Castle Logix items (blocks, towers, box and book). It was possible for researchers to use keyboard shortcuts to return these objects to their initial position on top of the table if participants dropped them. The interface included only visual cues as sensory modalities and there was no haptic nor audio feedback. The participants used their hands to manipulate blocks when building a castle. Overall, the system included HMD HTC Vive Pro, two Lighthouse tracking system base stations, and Leap Motion for hand tracking. The VR environment was developed with the Unity game development platform and it ran on the SteamVR game platform.

2.3.3. Augmented reality system

The AR system included the same physical wooden blocks as in the real test setup (Fig. 3). As well as the VR system, the AR system was made with Unity. The participants wore the Microsoft HoloLens HMD (Fig. 4), through which they saw the augmented outline of the castle and their real physical surroundings (Fig. 2). The participants used their hands to manipulate the real blocks, towers, and the book, but virtual buttons augmented onto the book were pressed using Microsoft HoloLens specific Air tap gestures. In the Air tap gesture the hand is held

Table 1
Virtual reality system description.

System model	Static model characteristics	• Castle blocks and towers • Book • Grid • Buttons on the book: 4 games, solution, return • Instruction texts: start, select, return, solution • Solutions for four castle options • Virtual hands • Castle Logix game box • Room: tables, chairs walls, windows, cabinets
	Dynamic model characteristics	• Hands are tracked and there are virtual models of the hands • Movable objects: blocks, towers, box, and book. These can be replaced at the initial positions if a participant manages to drop them onto the floor or chair
Interface	Sensory modalities	• Visual cues • No haptics, no audio
	Means of manipulation	• Participants use their hands to press virtual buttons and move virtual blocks, towers, box, and book
Technology	Virtual reality technologies (software and hardware)	• Head-mounted display HTC Vive Pro • Two base stations/lighthouses for tracking HTC Vive Pro • SteamVR • Unity was used for development of the virtual reality system • Blender was used for asset creation • Leap Motion was used for hand tracking

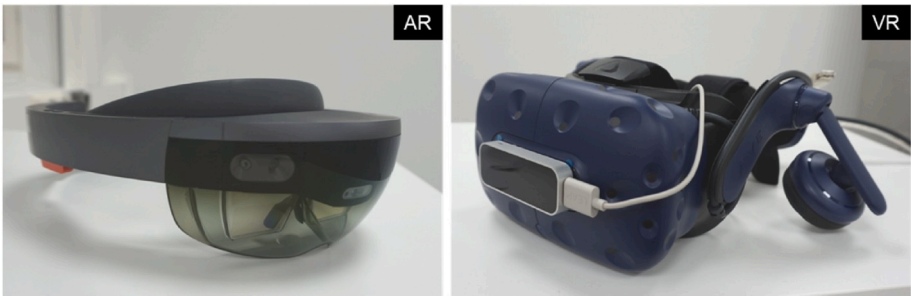


Fig. 4. Head-mounted displays used for augmented reality setup Microsoft HoloLens (left) and virtual reality setup HTC Vive with Leap Motion hand tracker on the front (right).

upright and the index finger is raised to the ready position. Then the finger is pressed down to tap or select and then back up to release, similar to a mouse click or select. For describing the AR system in detail, a systematic approach (Aromaa, 2017) was adopted (Table 2). 3D models in the AR system are a grid, castle's outlines, buttons on an instruction book and solutions for four different castles. The book's front and back covers were used for tracking for the grid placement and displaying the interface. In addition, there was a HoloLens cursor for interacting with the buttons and animated text above the buttons. The interface included visual cues and audio feedback for the button clicks as sensory modalities. For means of manipulation, the participants were able to use their hands to manipulate real blocks, towers and book, and air tap gestures to press virtual buttons. The system includes Microsoft HoloLens, Unity 3D, Microsoft Mixed Reality Tool Kit, Blender and Vuforia SDK.

2.4. Data collection methods

The data was collected with two approaches: user study and expert evaluation. The user study was executed in laboratory settings. In the expert evaluation, AR/VR experts answered a web-based questionnaire. The user study focussed on the experience of the real-world awareness and the expert evaluation positioned the used systems on the RV continuum to give a reference on how "virtual" the used AR system was. In addition, it is important to know if the AR experts have the same kind of understanding of the system's virtuality level with the users. This is essential when AR experts are developing systems to be used in industrial settings.

2.4.1. User study

Questionnaires, interviews, observations and video recordings were used as data collection methods. The study was designed based on the Finnish National Board on Research Integrity TENK guidelines for ethical research. The participants signed an informed consent form. The form included information of the project, research study, benefits and possible drawbacks, confidentiality of the data, data protection, personal data processing and rights of the participant. It was informed that the AR and VR technologies may cause nausea and disorientation for test

Table 2
Augmented reality system description.

System model	Static model characteristics	• Grid • Castle's outlines • Virtual Buttons on a book: 4 games, start, select, hint, return • Solutions for 4 castles
	Dynamic model characteristics	• Book's front and back covers are tracked for the grid placement and displaying the interface (start/restart/select/solution/return buttons)
Interface	Sensory modalities	• HoloLens cursor for interacting with the buttons • Text above the buttons is animated when the cursor is on the button • Visual cues • Audio feedback for the button clicks
	Means of manipulation	• Participants use their hands to manipulate the real blocks, towers, and book • Participants use HoloLens cursor and Air tap gesture to press virtual buttons
Technology	Virtual reality technologies (software and hardware)	• Microsoft HoloLens • Unity • Blender for asset creation • Vuforia SDK for tracking Image Targets (book's front and back covers)

participants. The research participants were also informed of their right to decline participating in the research and suspend or cancel their participation in the research at any time during the research without offering any reason for such action.

The participants' demographics were collected with a questionnaire. The real-world awareness questionnaire was developed based on several presence questionnaires (Dinh et al., 1999; Kim and Biocca, 1997; Schubert et al., 2001; Van Schaik et al., 2004). However, many of the questions were modified due to the challenge of using questions regarding presence in the context of both real and AR systems. According to Usuh et al. (2000) and Wagner et al. (2009), some VR-related presence questions in the real environment are interpreted as something else (e.g. involvement) and methods from VR presence research are not meaningful or informative into AR experiences. In this study, the real-world experience questionnaire had five questions, which were answered using a 5-point Likert scale (from 5 = strongly agree to 1 = strongly disagree). The questionnaire measured awareness, attention, noticing an approaching person, getting back to the game after distraction and captivation of the game. These aspects were chosen to give an understanding of the distraction AR systems may provide. In addition, after using the AR system, participants selected if they felt they were more within a real or virtual environment and defined their selection on the RV continuum (on a tick marks scale 0–21). An unweighted NASA-TLX (Hart and Staveland, 1988) questionnaire was used to collect the experience of subjective workload during the use of the systems. The NASA-TLX includes six subscales which assess mental demand, physical demand, temporal demand, effort, performance, and frustration level in a 21 vertical tick marks scale. In this study, the scale used was 0–100. The NASA-TLX questionnaire is useful when comparing different setups and their effect on the subjective workload. The experience of load is important to consider when applying AR systems to industry. In the end of the study, participants selected which system they liked the most, and were able to verbally express their overall experience and/or clarify their answers to the questionnaires.

2.4.2. Expert evaluation

A different questionnaire was made for the AR/VR experts. The questionnaire included screenshots and pictures of the AR and VR systems and descriptions of them. After the AR/VR experts were familiarized with the systems, they were asked to position the AR system on the RV continuum with two ways: first by evaluating the AR system's virtuality level (opinion 1) and by evaluating users' experiences of virtuality when using the AR system (opinion 2). The scale on the RV continuum was 0–11. The RV continuum scale was changed inadvertently from the user study (0–21) but this was taken into account when selecting data analysis methods (see later section, Data analysis methods). The experts were also asked to select either real or virtual environment where they estimated users felt being within while using the AR system.

2.5. Test procedure

The timeline of the test procedure is shown in Fig. 5. First, participants were informed of the study goals and the three test setups. Then, an informed consent form was signed and demographics were filled in. All participants performed castle building in the following order: real, VR and AR test setup. The purpose for the test order was to give participants an experience of both ends of the RV continuum, and then have an experience of the AR system in between these ends. This prepared them for positioning the AR system on the RV continuum when asked in the questionnaire. The same test order was selected even though it was known that when not using counterbalancing of different conditions it may lead to order effects. This decision was made because the participants were not requested to have previous experience of AR/VR technologies. It could have been difficult for a participant to actually evaluate the AR system's position on the RV continuum if they have not

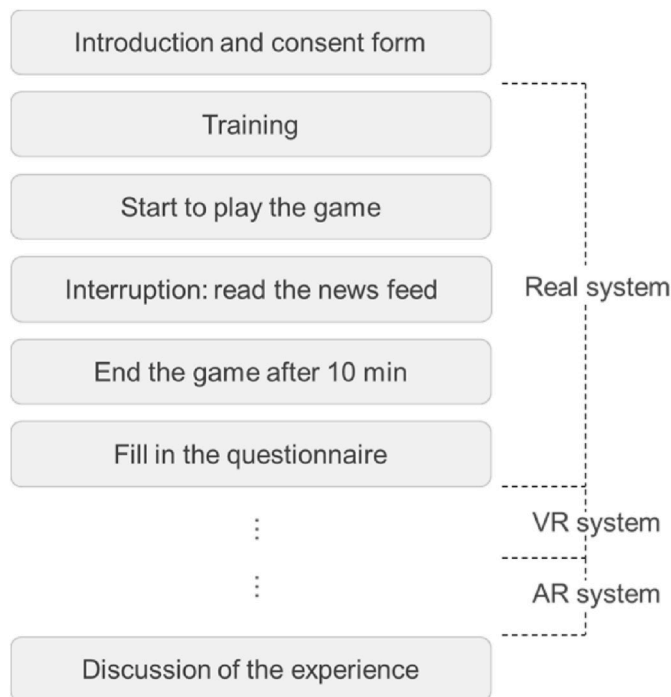


Fig. 5. Test procedure for the real, VR, and AR setups.

experienced its ends (real and virtual) before.

For each test setup, the participants received training and time to get familiar with the system just before the actual test (Fig. 5). In the real game training, the basic logic of the game was introduced and the participants solved an easy challenge, which meant that the participants were shown the complete solution including the blocks' colours and their placement. For the VR test setup training, the participants were able to try out picking and placing the Castle Logix blocks to get used to manipulation in that environment. Training for the AR test setup was conducted with the HoloLens Learn Gestures application. The participants learned how to use the Air tap gesture command. After each training, participants had 10 min to play the game with that particular system. During the game, the participants checked the instructions for building a castle. The instruction was an outline of a castle in a book (real case) or 3D outlines (VR and AR cases). After building a castle, the participants were able to check if the castle was built correctly. If it was correct, they were able to proceed to the next one. During the 10-min game playing period, the participants were interrupted intentionally after 5 min: a researcher walked from rear left of a participant and stopped next to them and verbally asked to read a news headline out loud from the researcher's smart phone. The researcher held the phone in front of the participant's field of view. Within the VR system, the participants raised the HMD to be able to read the headline. The interruption was used to understand participants' awareness of their surroundings and if they were only focusing on the virtual world objects (see for example, inattentive blindness, Simons and Chabris, 1999). The participants answered a questionnaire after the playing time in each setup. After performing all test setups, the participants selected which system they preferred and their experiences were discussed. They had a chance to verbally reason their answers or express their overall experience. On average, the duration of the test procedure for a participant was 1 h and 15 min. The participants were encouraged to build as many castles as they had time to but the number of the castles was not counted. The purpose was to make participants focus on the task but not take it as a competition.

2.6. Data analysis methods

Friedman test was used to test differences between the three types of setups because the observations were dependent (each type of system was assessed by all study participants) and measured in ordinal scale (in our study we used 5 point Likert scale). If the Friedman's test showed a statistically significant difference across the setups, the Wilcoxon signed-rank test was conducted to analyse the pair-wise differences with a Bonferroni correction in a significance level set at $p < 0.017$. In addition, the Mann-Whitney U test was used for comparison of experts' and users' opinions. Because the scales of RV continuum differed between users study (range: 0–21) and expert evaluation (range: 0–11) each observation was normalized prior analyses.

3. Results

This section describes the results from the user study and expert evaluation. Results of the real-world awareness and NASA-TLX questions in the three test setups are shown in Table 3.

3.1. Experience of the real-world awareness

There was a statistically significant difference in *perceived awareness* ("I was aware of the real-world surroundings while playing the game (i. e., sounds, room temperature, etc.)") depending on the type of system was used ($p < 0.017$). Wilcoxon signed-rank test showed that there was no significant difference between the real logic game block setup and AR setup ($Z = -0.973$, $p = 0.331$). However, there were statistically significant differences in perceived awareness in real vs VR setup ($Z = -3.236$, $p = 0.001$) and in AR setup vs VR setup ($Z = -3.082$, $p = 0.002$). Therefore, it can be said that the awareness of real-world surroundings was quite similar between the real and AR system but the awareness in the VR system differed from the real and AR system.

There was a statistically significant difference in *perceived attention* ("I paid attention to the real environment") depending on which type of system was used ($p < 0.017$). Post hoc analysis revealed that there were no significant differences between the real and AR setups ($Z = -0.237$, $p = 0.813$). There were statistically significant differences in perceived attention in the real setup vs VR setup ($Z = -3.018$, $p = 0.003$) and in the AR vs VR setup ($Z = -3.567$, $p < 0.001$). The participants paid attention to the real environment during real and AR setups more than in the VR test setup.

There was a statistically significant difference in *noticing an approaching person* ("I noticed the person who was bringing the smart-phone from far away") depending on which type of system was used ($p < 0.017$). Again, the real and AR setups did not differ statistically significantly ($Z = -1.904$, $p = 0.057$). There were statistically significant differences in notice the person in the real vs VR setup ($Z = -3.363$, $p = 0.001$) and in the AR vs VR setup ($Z = -2.724$, $p = 0.006$). The participants noticed the person in a similar way between real and AR conditions. The VR setup differed from these two.

There was no statistically significant difference in *resuming the task* ("It was easy to get back to the game after using the smartphone") after reading the newsfeed depending on which type of system was used ($p = 0.200$). The getting back to playing the game after reading a newsfeed was considered to be similar in each setup. In addition, there was no statistically significant difference of *how captivating* ("I was captivated by the game") the game felt depending on which type of system was used ($p = 0.254$).

Based on the results, it can be said that the experience of the AR system was quite near the experience of the real system. The experience of the VR system differed from the real and AR systems. It can be concluded that hypothesis H1 was not supported because the awareness in real and AR setups was similar and the participants did not experience that the AR system would reduce their real-world awareness.

Table 3

Median values, 25th and 75th percentiles, mean values and standard deviations of the real-world awareness and perceived workload (NASA-TLX) results for each test setup: real, augmented reality (AR) and virtual reality (VR). Results are presented here in the order of their reality-virtuality level, however, the test order was real, VR and AR (see Fig. 5). In addition, Friedman's test statistic and the corresponding p-value are described. The statements for each questionnaire are shown in the left column.

	Real		AR		VR		Friedman's test statistic and the corresponding p-value
	Median (25th-75th percentile)	Mean (SD)	Median (25th-75th percentile)	Mean (SD)	Median (25th-75th percentile)	Mean (SD)	
Real-world awareness							
I was aware of the real-world surroundings while playing the game (i.e., sounds, room temperature, etc.)	4 (4–5)	4.1 (1.1)	4 (3–5)	3.8 (1.1)	3 (2–4)	2.9 (1.2)	$\chi^2(2) = 18.808, p < 0.017$
I paid attention to the real environment	4 (3–5)	3.7 (1.0)	4 (3–4)	3.6 (0.8)	2 (1–3)	2.3 (1.2)	$\chi^2(2) = 20.455, p < 0.017$
I noticed the person who was bringing the smartphone from far away	4 (3–4)	3.8 (0.9)	4 (2–4)	3.3 (1.1)	2 (1–4)	2.4 (1.3)	$\chi^2(2) = 18.107, p < 0.017$
It was easy to get back to the game after using the smartphone	5 (4–5)	4.4 (0.8)	4 (4–5)	4.1 (0.9)	4 (4–5)	4.1 (1.0)	$\chi^2(2) = 3.217, p = 0.200$
I was captivated by the game	4 (4–5)	4.3 (0.7)	4 (3–5)	3.7 (1.2)	4 (3–5)	3.8 (1.4)	$\chi^2(2) = 2.792, p = 0.254$
Perceived workload (NASA-TLX)							
Mental Demand	65 (40–80)	59.7 (22.9)	60 (35–65)	53.2 (19.1)	70 (50–80)	61.6 (23.0)	$\chi^2(2) = 2.563, p = 0.300$
Physical Demand	10 (5–20)	14.2 (16.9)	30 (15–50)	37.4 (27.5)	50 (25–85)	53.7 (29.1)	$\chi^2(2) = 23.743, p < 0.017$
Temporal Demand	40 (20–50)	36.6 (20.1)	25 (10–50)	29.5 (21.5)	30 (15–50)	33.9 (21.2)	$\chi^2(2) = 4.212, p = 0.124$
Performance	25 (15–35)	32.4 (23.6)	20 (10–35)	29.2 (26.2)	80 (50–90)	70.0 (27.8)	$\chi^2(2) = 21.444, p < 0.017$
Effort	50 (40–75)	55.5 (21.0)	40 (25–65)	46.3 (23.6)	85 (65–90)	75.5 (21.4)	$\chi^2(2) = 16.829, p < 0.017$
Frustration	30 (15–60)	35.0 (24.4)	50 (20–65)	42.6 (27.3)	70 (50–85)	62.9 (28.1)	$\chi^2(2) = 13.971, p = 0.001$

3.2. Perceived workload

In the NASA-TLX questionnaire, *mental demand* and *temporal demand* did not differ between the type of systems used, $p = 0.3$ and $p = 0.124$, respectively.

There was a statistically significant difference in perceived *physical load* depending on which type of system was used ($p < 0.017$). Post hoc analysis revealed that there were no significant differences between the AR and VR setups ($Z = -2.204, p = 0.028$). There were statistically significant differences in perceived physical load in the real vs VR setup ($Z = -3.270, p = 0.001$) and in the real vs AR setup ($Z = -2.987, p = 0.003$). The task was physically demanding when using HMDs (AR and VR systems) compared to the real test setup without any wearables.

There was a statistically significant difference in perceived *performance* depending on which type of system was used ($p < 0.017$). Post hoc analysis revealed that there were no significant differences between the real and AR setups ($Z = -0.597, p = 0.550$). There were statistically significant differences in perceived performance in the real vs VR setup ($Z = -3.100, p = 0.002$) and in the AR vs VR setup ($Z = -3.386, p = 0.001$). The participants did not feel that they were able to perform the task successfully when using the VR system.

There was a statistically significant difference in perceived *effort* depending on which type of system was used ($p < 0.017$). Post hoc analysis revealed that there were no significant differences between the real and AR setups ($Z = -1.110, p = 0.267$). There were statistically significant differences in perceived effort in the real vs VR setup ($Z = -3.250, p = 0.001$) and in the AR vs VR setup ($Z = -3.151, p = 0.002$). The participants needed to work harder with the VR system to accomplish the level of performance.

There was a statistically significant difference in perceived *frustration* depending on which type of system was used ($p = 0.001$). Post hoc analysis revealed that there were no significant differences between the real and AR setups ($Z = -0.850, p = 0.398$). There were statistically significant differences in perceived effort in the real vs VR setup ($Z = -3.270, p = 0.001$) and in the AR vs VR setup ($Z = -3.109, p = 0.002$).

The participants felt frustrated with the VR system.

There is no threshold level in the NASA-TLX for high workload (Hart and Staveland, 1988). However, some suggestions for the general workload thresholds have been proposed, for example, 40 ± 10 (out of 100) (Colle and Reid, 2005), 50–55 (Mazur et al., 2014, 2013) and over 50 (Yu et al., 2015). In the real system, mean values of mental demand and effort were over the threshold (50) level. This suggests that the game required quite much thinking and concentration by the participants for them to be able to build castles. Within the VR system, the mean values of mental demand, physical demand, performance, effort and frustration were all over the threshold level. As such, the system has a high workload and could not be used for a long period of time. The AR system had only one mean value over the threshold level: mental demand.

Overall, it can be concluded that the perceived workload for the VR system was high and for the real and AR system it was rather low or moderate. This suggests that the VR system differed quite much from the real and AR systems when considering the workload.

3.3. Experience on the reality-virtuality continuum

The participants selected in which environment they felt most immersed while using the AR system. 16 (84%) participants felt more immersed in the real environment and 3 (16%) felt more immersed in the virtual environment. When experts were estimating users' feelings regarding this same question, 9 (82%) of the experts selected that users felt more immersed in the real environment and 2 (18%) in the virtual environment. The participants also defined their feelings about virtuality on the RV continuum. The experts were also asked to define their opinion of the AR system's technology in the Milgram's RV continuum. Mann-Whitney U test showed that there was no significant difference between experts' and test participants' opinions ($U = 98, p = 0.790$) about the AR system's position in the RV continuum (Opinion 1, Fig. 6). In addition, the experts were asked to mark in the RV continuum the experience they believe users had when using the AR system. This was compared with the users' selection again. Mann-Whitney U test revealed

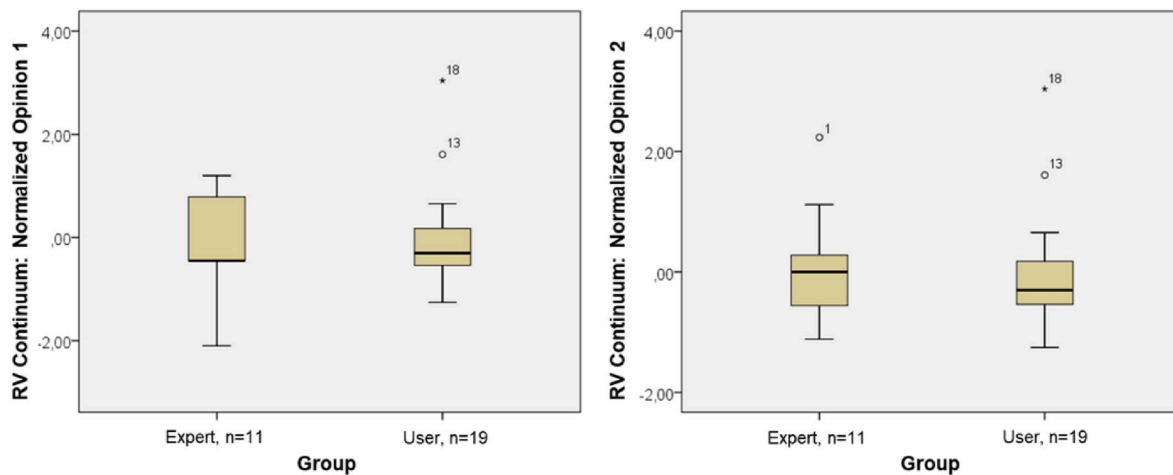


Fig. 6. Box plot of normalized opinions of expert and participant groups in reality-virtuality (RV) continuum. In both plots there is users' experience of virtuality level when using the AR system. On the left (Opinion 1), the experts' opinion of the AR system in the reality-virtuality continuum is compared with the users' experience. On the right (Opinion 2), the experts' expectations of the users' experience of the virtuality level is compared with the users' experience.

that there was no significant difference between experts' estimation of user's experience and test participants' experience ($U = 97$, $p = 0.758$) of the AR system's position in the RV continuum (Opinion 2, Fig. 6).

3.4. Participants' preferences

In the end of the study, the participants were asked which of the game setups they liked the most. 11 (58%) participants liked the real system and 8 (42%) participants liked the AR system. Even though none of the participants chose the VR system some of them said that the VR HMD was more comfortable to use than the HMD in the AR system. In addition, one mentioned that they would first choose the real system, followed by the VR system and then AR. They claimed that the AR system required much effort and use of poor postures due to the small field of view. There were also comments stating that the AR HMD was heavy, their neck hurt afterwards and a couple participants got headache symptoms from AR because they wore their eye glasses under the AR HMD. However, one said that the AR HMD was better than VR HMD after setting it right. Many participants agreed that the VR system was difficult to use because there was no haptic feedback and fingers easily went inside the virtual blocks. In addition, the participants claimed that the usability of the VR system should have been better.

4. Discussion

The purpose of this study was to understand how distractive the use of HMD-based AR system is. The use of AR system was compared to real and virtual setups to understand real-world awareness, perceived workload and the experience in the RV continuum. In addition, implications to the user studies are suggested and limitations and future work are addressed.

4.1. Experience of the real-world awareness, perceived workload and experience on the reality-virtuality continuum

Based on the results it can be said that the use of this kind of HMD-based AR system was quite near to the experience of the real system when considering attention- and awareness-related issues. For example, the participants noticed the person who was bringing the smartphone in a similar way between real and AR setups. Therefore, wearing the HoloLens HMD in this kind of task does not decrease awareness of real-world surroundings. This may have been due to the low level of virtual objects and interactivity of the AR system: most of the time participants manipulated real blocks of the game, which was equal to the real case

scenario. In addition, the HoloLens device is based on a see-through solution so the participants are able to visually see their environment. Regardless, the results show that the use of an HMD-based AR system did not reduce real-world awareness, which would lend support to applying AR systems in industrial contexts.

This study was, to the best of our knowledge, the first attempt to understand the real-world awareness of users when comparing real, AR and VR setups using HMDs. However, the topic requires more research with AR systems that are more interactive and include more virtual objects, and are tested in real industrial environments. For example, this study could be repeated with less physical tangible objects and with more virtual content to evaluate the effect on the experience of real-world surroundings. In addition, the distraction happened only once during the study and the participants were not asked to particularly pay attention to their environment (as they would have been, for example on a factory floor). The distraction happened always after 5 min of playing the game and it was possible that the participants were just building the castle with physical objects and not using the virtual content at that time. Therefore, the experience was quite similar with the real test setup.

Based on the NASA-TLX physical demand scale, the participants felt that the use of (VR and AR) HMDs is more demanding than performing a task without wearables. The participants also verbally commented that wearing HMDs was still somewhat uncomfortable. Aaltonen and Laarni (2017) agree that HMDs can be problematic for comfort. In addition, Masood and Egger (2019) say that ergonomics (e.g. weight of the HMDs) is important for successful implementation of AR systems for industry use. The median and mean on physical demand with the VR system was higher than for the AR system. This indicates that the participants had also other load issues, for example due to poor usability. Kim et al. (2019) acknowledge that job performance, workload, and usability were more affected by user interface designs than HMD type. In addition, in our study half of the participants had eyeglasses and some of them wore glasses under the HMDs. This could also affect the level of physical demand because it is possible that the eyeglasses do not fit well under the HMDs or leaving them out may cause eyestrain. Palmari et al. (2018) agree that HMDs have to become more comfortable.

Based on NASA-TLX results in performance, effort and frustration, the participants felt that the VR system was not very good. The VR system was significantly harder to use and the usability of the user interface was poor. At the end of the study, none of the participants selected the VR setup as the game they liked the most. However, some of the participants said that they would have chosen it if the usability was improved. The reason for this was that the Leap Motion tracking system lost the hand tracking occasionally and the participants were not able to

manipulate objects and proceed in tasks fluently. Sometimes the Leap Motion did not track the fingers because they were behind the palm. In addition, the tracking system had difficulties in recognizing hands when they were on top of the white table. It was also learned that the system was sensitive to room light reflections from the table and the light-coloured table did not offer enough contrast with the hands. In addition, some of the participants commented that it was difficult to manipulate objects without haptic feedback. They observed that their fingers went inside the virtual blocks.

When studying medians and means of NASA-TLX scores, it was seen that the real setup was slightly more demanding than the AR setup in mental demand, temporal demand, performance and effort. Regarding preferences, however, the real system was slightly preferred over the AR system. This could be due to learning effect: in the test trials the real test setup was always the first one. It seems that after playing the game several times, the participants became more familiar with the logic of the game. Towards the end they did not feel so much mental demand from performing well in the game. In addition, the castle challenges were not as demanding within VR and AR systems compared to the real-life setup. By randomizing the order of the setups and/or training the participants more beforehand could have decreased the learning effect.

The nature of the task has a major role in the evaluation of the awareness of the real-world surroundings when using an AR system. In this study, a simple but captivating game was used on top of the table while being seated. In this setup, the participants did not need to pay attention to the real-world surroundings, did not need to walk around or be aware of the traffic nearby. However, there are tasks in industry work which can be quite different. For example, if a person is doing a maintenance task on the factory floor where there may be a lot of forklift traffic or on the top of an industrial crane, it is important they to be aware of the surroundings. Different maintenance operations can have different AR system requirements (Palmarini et al., 2018). In addition, the AR content has an effect on the users' awareness: the amount of content, visual appearance, dynamics, etc. In this study, the static castle outlines were projected to the participant's field of view so they would not need to turn their head and change the focus. The participants experienced the game similarly captivating in real, VR and AR games (there was no significant difference in responses to this question). These results suggest that the nature of the game was very well reproduced for AR and VR systems.

Experts' opinions of the AR system's virtuality level on the RV continuum was similar to the participants' experiences. This suggests that with the low-level virtuality, AR developers have a good understanding of the user experience. The RV continuum is created for categorising displays, but based on this study with the "low-virtuality-level" AR systems, it seems to correlate also with the user experience of the virtuality level. Therefore, the RV continuum could predict also users' experience of the virtuality. However, this should be studied more and tested with systems including more virtual objects. The use of the estimates on the virtuality level could be used in future field studies to estimate how far along the RV continuum an AR system can be taken without the users, or industrial workers, degrading their real-world awareness to a level endangering their safety.

4.2. Implications to the user studies when evaluating awareness of real-world surroundings when using AR systems

While comparing the three test setups (real, VR, AR) it was difficult to find questionnaires that would be applicable for all three of them. The term "presence" in real, VR and AR means different things: in real life a user is present in the real environment, in VR the user feels a presence in the virtual environment (and the aim is that the user forgets the real environment), and presence in AR is difficult to interpret (the user needs to be present in the real world, but their focus might be on virtual objects). Usoh et al. (2000) and Wagner et al. (2009) agree that current presence questions are difficult to adapt to real and AR systems. They

say that some VR-related presence questions when used in a real environment are interpreted as something else (e.g. involvement). In addition, methods from presence research related to VR are not meaningful or informative in AR experiences. Our real-world awareness questions were applicable for the comparison of real, VR and AR test setups. They were understood very well by the participants in each test setup.

There seems to be a lack of methods for considering attention and awareness issues when studying AR systems' distraction from the real-world surroundings. There are many studies related to the usability of AR systems, but there are no commonly approved methods for evaluating the distraction of HMD-based AR systems. For example, there is a standard method for testing operators' field of view in earth-moving machinery (SFS-ISO 5006, 2017), which describes the test method and performance criteria. In our study, a person with a mobile phone distracted the participants. This was only one distraction during the performance. It would be important to consider in more detail which kind of distractions there should be and how they should be performed when considering the field of view of participants.

When doing user studies with AR/VR systems it is always difficult to generalize conclusions because results are always very dependable on the used AR/VR systems. As we saw in our study, the AR system did not have lot of virtual objects and did not require constant interaction with the virtual objects: the participants said that the experience was quite near with the real system experience. However, if the system would have included more virtual objects and the participants would have interacted with them actively, the experience of the awareness and attention might have been quite different. Due to this challenge, it is important that the used AR/VR systems are described in detail and systems' positions are estimated on the RV continuum, because small differences in AR/VR systems may provide different results.

Some AR/VR technologies are not mature enough and therefore, test setups which evaluate AR/VR systems can be quite novel. In this study, the hand tracking system failed to be consistent enough and made participants frustrated with the VR system. This may create challenges for user studies if the usability of the system is not at a good level.

When studying AR systems, it might be challenging to ensure that users see augmented information in the right way and in the right place. When studying VR systems, it is easy to duplicate the participants' view to a display for researchers to see. With an HMD-based AR system it was difficult to actually know what the participants are seeing. Therefore, it is possible that the participants would adopt the incorrect use practices of AR system which may lead, for example, to poor postures.

4.3. Study limitations and future research

Some limitations may have had an effect on the validity of the research. The counterbalancing of different conditions was not used because the initial purpose of the study was to set the experience of AR system on the RV continuum: the participants were not required to have earlier experience of AR/VR technologies and therefore they were given first experiences from both ends of the RV continuum (real and virtual) to be able to position the AR experience on the continuum. Due to this ordering, learning might have happened during the study and therefore the participants may have experienced that the last system used was not so mentally demanding. It can be seen that the study could have benefitted of the counterbalancing but it was not possible to repeat the study due to financial and scheduling issues. Nevertheless, this study creates new knowledge related to users' real world awareness when using AR systems and can be seen introductory research to initiate and direct studies in this field.

The VR system was experienced as quite uncomfortable and difficult to use and it may have distracted the participant from the study's goals. In addition, only 11 AR/VR experts answered the web questionnaire. The low number of the experts may have affected the results.

The study was based on subjective measures but in addition, objective measures could have enhanced the results. Eye-tracking is one of the

most prominent approach to use when studying awareness, distractions and their effects on the user. Various physiological measures such as pulse and galvanic skin response could be used, if the person's activity and stress levels were of interest. The time to resume to the task could be one measure also, but in this task it was difficult to implement: some of the participants held blocks in their hands when reading the news title and got back to task really quickly. In addition, in the VR task the participants were required to lift their HMD off which took more time. In future studies, there could be more tasks and different kind of distractions at randomized intervals, for example, more visual cues to participants to see in their field of view or other people/traffic passing by. But the challenge of creating the same test setup for virtual environment and AR remains. In addition, it is suggested to test awareness in a factory look-alike laboratory which replicates the actual factory environment.

This is a novel study that addresses the experience of the AR systems in the RV continuum. The study addressed, especially, the distraction AR systems may create. In the future, the same study should be replicated with a different level of virtual content and interactivity in AR systems. Thus, it would be possible to study in more detail which kinds of AR content distract the users and if the users experience the AR systems always in the same way. It would be also interesting to study the linkage between workload and real-world awareness. In addition, we would like to develop further methods and questionnaires for how to evaluate distraction with HMD-based AR systems.

5. Conclusions

The use of AR systems is increasing in industry. However, there are still some concerns that AR systems may distract the user and create safety risks. The purpose of this study was to understand if the use of a HMD-based AR system distract users' awareness of the environment. The experience was evaluated in a laboratory study by comparing three different test setups: real system, AR system and VR system.

Results indicate that an HMD-based AR system with a moderate level of virtual content can be seen as positioned near the real-world situation. The participants were aware of the surroundings, and virtual objects did not represent remarkable distractions. Although, it should be noted that the participants were sitting at a desk and they did not carry out any demanding work task in a safety-critical environment. In addition, the participants agreed that the comfort of the HMDs could be improved. The participants' experience and the experts' estimation of the AR system's location in the virtuality-reality continuum were similar.

This novel study addressed the topic of how much HMD-based AR systems distract users. In the future, there should be studies in which AR systems include more virtual content and which are tested with more distractions in a real industrial environment. During the study, it was noticed that the evaluation of the AR system and comparison with real and VR systems is challenging. For this reason, methods addressing attention and awareness issues when using AR systems are needed. The findings of the study can provide guidance for designing and testing AR systems, and implementing AR systems in industry.

Declaration of competing interest

None.

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